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RESEARCH ARTICLE

Impact of Chemical Reaction Over A Stretching Cylinder Embedded In A Porous Medium With Lorentz Forces

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ABSTRACT:

An analysis has been carried out to describe the impact of presence of magnetic field and mass transfer of viscous fluid over a horizontal stretching cylinder embedded in a porous medium with chemical reaction. The problem is modelled mathematically and the governing partial differential equations are converted into non-linear differential equations by using suitable similarity transformations. The resulting non-linear coupled differential equations are solved numerically by using fourth order Runge-Kutta scheme together with shooting method. The impact of significant parameters on velocity, temperature, concentration, skin friction coefficient and the Nusselt number has been studied and numerical results are presented graphically and in tabular form.

KEYWORDS: Stretched cylinder surface, MHD, mass transfer, chemical reactions, porous medium.

INTRODUCTION:

The boundary layer flow of heat and mass transfer induced by stretching surfaces have many applications in the process of extrusion and also have importance in theory. Applications of such flows occur in aerodynamics, drawing of plastic sheets, paper productions, cooling of metallic plates etc. The heat transfer phenomenon takes places due to the temperature difference between two objects. However, analysis of flow and heat transfer phenomena over stretching cylinder is also important in processes like fiber and wire drawing, hot rolling etc. Based on these applications, Crane [1] initiated the study on boundary layer flow induced due to a stretching cylinder.

The work which was done by Crane have extended by Wang [2] to examine the flow and heat transfer outside a hollow stretching cylinder. Ishak and Nazar [3] numerically investigated the heat transfer characteristic in viscous fluid flow over a stretching cylinder.

Convection flows with heat and mass transfer in a porous medium have gained significant attention in the recent years because of their engineering applications such as store of nuclear waste materials ,geothermal systems, oil extraction, solid matrix heat exchangers, and thermal insulations.

The other application include ground water hydrology, underground coal gasification, energy efficient drying processes and natural convection in earth's crust. In a mixed convection problem, when the free stream velocity of the fluid is small and the if there is large difference in temperature and concentration between the surface and ambient fluid, then the Buoyancy effects on forced convective heat and mass transfer become significant. A comprehensive reviews on this topic have been given in the books by Ingham and Pop [4, 5] Nield and Bejan [6], Vafai [7] and Pop and Ingham [8]. The axisymmetric laminar boundary layer flow of a viscous incompressible fluid and heat transfer towards a stretching cylinder embedded in porous medium was studied by Mukhopadhyay [9]. Very recently, the influences of the temperature-dependent fluid properties and the transverse curvature on the axisymmetric flow and heat transfer at a horizontal permeable stretching

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cylinder were explored by Vajravelu et al. [10]. The mixed convective flow along a stretching cylinder embedded in a thermally stratified medium was examined by Mukhopadhyay and Ishak [11]. Shateyi and Marewo [12] made use of SRM (successive relaxation method) method to discuss the axisymmetric flow over a stretching cylinder embedded in a porous medium. Si et al. [13] presented the numerical solution of unsteady flow and heat transfer of viscous fluid due to porous cylinder which is stretching and expanding at the same time.

Magnetohydrodynamics (MHD) is the study of the interaction of conducting fluids with electromagnetic phenomena which has important applications in industrial fields. For example, equipments such as MHD generators, flow meters, bearings, pumps and boundary layer control are affected by the interaction between the electrically conducting fluid and a magnetic field. MHD problems occur in several situations like the prediction of space weather, damping of turbulent fluctuations in semiconductor melts in crystal growth, measurement of flow rates of beverages in food industry. Hydromagnetic boundary layers are observed in several technical systems employing liquid metal and plasma flow transverse of magnetic fields. In these cases, the flow control can be realized by the Lorentz force. Recently, many publications discussed the effect of magnetic field on the boundary layer flow problems along a stretching sheet in transverse direction [14–16]. Bala Anki Reddy et.al [17] examined the numerical Study of MHD Boundary Layer Slip Flow of a Maxwell Nanofluid over an exponentially stretching surface with convective boundary condition. Also found that the thermal boundary layer thickness increases with Brownian motion and thermophoresis parameters. Bala Anki Reddy and Bhaskar Reddy [18] analyzed thermal radiation effects on an exponentially stretching sheet with MHD.

In many chemical engineering processes, chemical reactions take place between a foreign mass and the working fluid which moves due to the stretching of a surface. During a chemical reaction heat is generated between two species (Byron Bird R. et al. [19]). The order of the chemical reactions depends on several factors. One of the simplest chemical reactions is the first-order homogeneous reaction in which the rate of reaction is directly proportional to the species concentration (Cussler [20]). The addition or diffusion of chemical reaction in the boundary layer flow has many

applications in water and air pollutions, atmospheric flows, fibrous insulation and other chemical engineering problems. The diffusion of a chemically reactive species in a laminar boundary layer flow was discussed by Chambré and Young [21]. Andersson et al. [22] investigated the transfer of a chemically reactive species over a linearly stretching sheet for homogenous first and higher order reactions. Takhar et al. [23] studied the flow and mass transfer on a stretching sheet with a magnetic field and chemically reactive species, while Raptis and Perdakis [24] considered the viscous flow in the presence of a chemical reaction and magnetic field over a nonlinearly stretching sheet. Muhaimin et al. [25] studied the effect of chemical reaction, heat and mass transfer on nonlinear magnetohydrodynamic (MHD) boundary layer past a porous shrinking sheet in the presence of suction. Patil et.al [26] investigated the chemical reaction effects on unsteady mixed convection boundary layer flow past a permeable slender vertical cylinder due to a nonlinearly stretching velocity. Bala Anki reddy [27] considered the steady two-dimensional MHD convective boundary layer flow of a Casson fluid over an exponentially inclined permeable stretching surface in the presence of thermal radiation and chemical reaction. Bala Anki Reddy and Raja Das [28] discussed the prediction of the magnetohydrodynamic boundary layer slip flow over a permeable stretched cylinder with chemical reaction.

Motivated by the above mentioned studies, the aim here is to investigate the impact of Chemical Reaction over a stretching cylinder embedded in a porous medium with Lorentz Forces. Mathematical modelling is presented here. The present study is structured in the following fashion. The mathematical formulation is completed in the next section. Series solutions by using a shooting technique with Runge–Kutta method of order four are developed in the subsequent sections. Then convergence analysis and discussion are presented. Important results are summarized in the last section. To the best of author's knowledge, such study has not been reported earlier in the literature.

MATHEMATICAL FORMULATION:

Consider a steady, axisymmetric mixed convection flow of viscous, incompressible electrically conducting fluid due to continuous stretching of a horizontal cylinder having radius a embedded in a porous medium in the presence of porous medium.

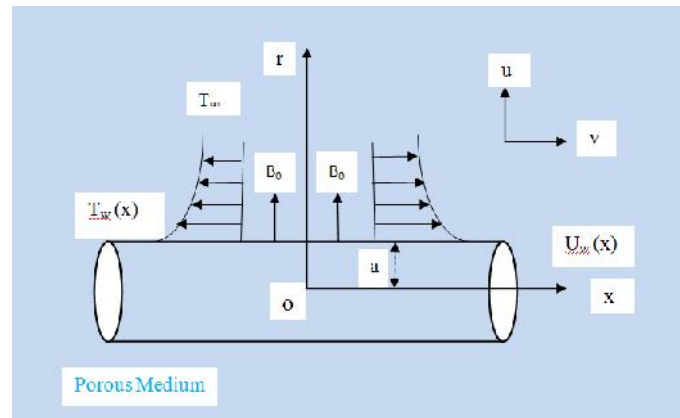


Fig. 1. Schematic representation of the considered problem

As presented in Fig.1, the direction of the x-axis is taken along the axis of the cylinder and the r-axis is considered to be in the radial direction. A uniform magnetic field of strength B_0 is applied in the radial direction, i.e., normal to the stretching cylinder.

The stretching velocity $U_w(x)$ of the cylinder is assumed to be of the form $U_w(x) = U_0 \left(\frac{x}{L} \right)$, where $U_0 > 0$ is a constant and L is the characteristic length. The surface of the stretching cylinder is subjected to a prescribed surface temperature $T_w(x) = T_\infty + T \left(\frac{x}{L} \right)^n$, the ambient fluid temperature is T_∞ , $C_w(x) = C_\infty + C_0 \left(\frac{x}{L} \right)^p$ is the prescribed surface concentration. The induced magnetic field effects are considered to be negligible due to which magnetic Reynolds number is not taken. Moreover, it is supposed that the viscous dissipation and joule dissipation effects are present. Under all these assumptions, the boundary layer equations that govern the flow and heat transfer phenomenon are written as:

$$\frac{\partial}{\partial x}(ru) + \frac{\partial}{\partial r}(rv) = 0 \quad (1)$$

$$u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial r} = \frac{\nu}{r} \frac{\partial}{\partial r} \left(r \frac{\partial u}{\partial r} \right) - \frac{\nu}{K_p} u - \frac{\pm B_0^2}{\sigma} u + g(T - T_\infty) + g^*(C - C_\infty) \quad (2)$$

$$u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial r} = \frac{k}{\rho C_p r} \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right) + \frac{\nu}{C_p} \left(\frac{\partial u}{\partial r} \right)^2 + \frac{\pm B_0^2}{\sigma C_p} u^2 + \frac{\nu}{C_p K_p} u^2 \quad (3)$$

$$u \frac{\partial C}{\partial x} + v \frac{\partial C}{\partial r} = \frac{D}{r} \frac{\partial}{\partial r} \left(r \frac{\partial C}{\partial r} \right) - \Gamma(C - C_\infty) \quad (4)$$

and the associated boundary conditions are:

$$u = U_w(x), v = 0, T = T_w(x), C = C_w(x) \quad \text{at } r = a$$

$$u \rightarrow 0, T \rightarrow T_\infty, C \rightarrow C_\infty \quad \text{at } r \rightarrow \infty \quad (5)$$

where (u, v) are the velocity components in the (x, r) directions, ν is the kinematic viscosity, ρ represents the fluid density, σ is the electrical conductivity of the fluid, B_0 is the uniform magnetic field, K_p is the permeability of the medium, k is the thermal diffusivity of the fluid, c_p is the specific heat at constant pressure, n is the temperature exponent, p is the concentration exponent and T is the temperature of the fluid.

Introducing the following similarity variables

$$\eta = \frac{r^2 - a^2}{2a} \left(\frac{U_w}{\nu x} \right)^{\frac{1}{2}}, \quad \psi = (U_w \nu x)^{\frac{1}{2}} f(\eta), \quad \theta = \frac{T - T_\infty}{T_w - T_\infty}, \quad \varphi = \frac{C - C_\infty}{C_w - C_\infty} \quad (6)$$

where η is the similarity variable, ψ denotes the stream function, f , θ and φ are the dimensionless stream

function, temperature and concentration respectively. The stream function is defined as

$$u = \frac{1}{r} \frac{\partial \psi}{\partial r} \text{ and } v = \frac{1}{r} \frac{\partial \psi}{\partial x} \quad (7)$$

By substitution of (6) and (7), the Eq. (1) is automatically satisfied and Eqs. (2)-(4) are converted into following nonlinear ordinary differential equation

$$(1 + 2\gamma_1) f''' + 2\gamma f'' + f f'' - f'^2 - (M + K) f' + \alpha'' + \phi \alpha = 0 \quad (8)$$

$$(1 + 2\gamma_1) \theta'' + 2\gamma \theta' + Pr (f' \theta - n f \theta') + Pr Ec (1 + 2\gamma_1) f'^2 + Pr Ec (M + K) f'^2 = 0 \quad (9)$$

$$(1 + 2\gamma_1) \alpha'' + 2\gamma \alpha' + Sc (f \alpha' - p f' \alpha - \alpha) = 0 \quad (10)$$

and the boundary conditions (5) upon introducing (6) becomes

$$\begin{aligned} f(0) &= 0, f'(0) = 1, \theta(0) = 1, \alpha(0) = 1 \\ f' &\rightarrow 0, \theta \rightarrow 0, \alpha \rightarrow 0 \text{ as } \eta \rightarrow \infty \end{aligned} \quad (11)$$

where the prime denotes differentiation with respect to η , $\gamma = 1/a(vL/U_0)^{1/2}$ is the curvature parameter, $M = \pm B_0^2 L / \mu U_0$ is the magnetic field parameter, $K = \mu L / U_0 K_p$ is the permeability parameter of the porous medium, $\alpha = \frac{g T_0 L^2}{U_0^2}$ is the Grashof number, $\phi = \frac{g^* C_0 L^2}{U_0^2}$ is the solutal Grashof number, $Pr = \mu / \kappa$ is the Prandtl number, $Sc = \mu / D$ is the Schmidt number, $\Gamma = \Gamma L / U_0$ is the chemical reaction parameter and $Ec = U_w^2 / c_p (T_w - T_\infty)$ is the Eckert number.

It is noticeable that when $\gamma = 0$ (flat surface) and $M = 0$, $K = 0$, the Eq. (8) reduces to that of Crane [27] which has the solution $f(\eta) = 1 - e^{-\eta}$. Moreover, also when $Ec = 0$, the considered problem is reduced to that of Grubka and Bobba [28].

The quantities that are of significant physical importance are skin friction coefficient, heat transfer rate and mass transfer and are defined as

$$C_f = \frac{2 \tau_w}{\frac{1}{2} \rho U_w^2}, \quad Nu_r = \frac{r q_w}{k (T_w - T_\infty)}, \quad Sh_r = \frac{x J_w}{D (C_w - C_\infty)} \quad (12)$$

Here $\tau_w = \mu \left(\frac{\partial u}{\partial r} \right)_{r=a}$ is the shear stress at the cylinder surface, $q_w = -k \left(\frac{\partial T}{\partial r} \right)_{r=a}$ is the surface heat flux, and $J_w = -D \left(\frac{\partial C}{\partial r} \right)_{r=a}$ is the surface mass flux. On plugging values of τ_w , q_w and J_w into (10) and making use of dimensionless variables (5), we get the non-dimensional form of skin friction coefficient, local Nusselt number and Sherwood number as

$$\frac{1}{2} Re_x^{1/2} C_f = f''(0), \quad Re_x^{-1/2} Nu_x = -\theta'(0), \quad Re_x^{-1/2} Sh_x = -\alpha'(0) \quad (13)$$

where $Re_x = U_w x / \nu$ is the local Reynolds number.

METHOD OF SOLUTION :

The set of non-linear ordinary differential equations (16) – (18) with boundary conditions (19) have been solved by using the Runge - Kutta fourth order method along with shooting technique. The step size $\Delta \eta = 0.01$ is used while obtaining the numerical solution with $\eta_{\max}$, and accuracy to the fifth decimal place is sufficient for convergence. A step size of $\Delta \eta = 0.01$ was selected to be satisfactory for a convergence criterion of 10^{-6} in nearly all cases. The maximum value of η_{∞} to each group of parameters M , K , Pr , Ec , n , Sc , p and Γ determined when the values of unknown boundary conditions at $\eta = 0$ not change to successful loop with error less than 10^{-6} .

RESULTS AND DISCUSSION:

Numerical results for different values of thermo physical parameters controlling the dynamics in the flow regime have been computed and presented in Figs. 2-8. The thermo physical parameters considered in this study include: Magnetic field parameter (M), curvature parameter (γ), Permeability parameter of the porous medium (K), Chemical reaction parameter (β), Prandtl number (Pr), Schmidt number (Sc), measure of the strength of the homogeneous reaction (K), Temperature exponent (n), and concentration exponent (p).

Fig. 2 illustrates the effects of curvature parameter γ and magnetic field parameter M on velocity profile $f'(\eta)$. The solid lines represent the case when there is no curvature and the stretching surface is flat whereas the dotted lines represent the problem of stretching cylinder in which the value of curvature parameter is equal to 2. From the figure, it is quite evident that the boundary layer effects last for longer distance in case of stretching cylinder as compared to flat stretching surface. The thickness of the boundary layer is more for curved cylinder as compared to flat stretching sheet. Further- more, it is noticed that increase in the value of magnetic field parameter M results in reduction of fluid velocity in both the cases. This is quite compatible with the physics of the problem as increase in M causes the Lorentz force to become strong which resists the fluid motion and slows it down. Fig. 3 demonstrates the effects of permeability parameter K on velocity profiles for flat stretching surface ($\gamma = 0.0$) and stretching cylinder having the value of curvature parameter equal to 2, i.e. $\gamma = 2.0$. In both the cases, increase in the value of permeability parameter enlarges the resistance caused by the porous medium to fluid flow which depreciates the fluid velocity.

The impact of increasing the Eckert number Ec on temperature $\theta(\eta)$ is illustrated in Fig. 4. An augmentation in the temperature and hence in the thermal boundary layer thickness is noticed for both the values of curvature parameter i.e., for $\gamma = 0.0$ which corresponds to flat stretching surface and for $\gamma = 2.0$ which represents a particular case of stretching cylinder. This increase in temperature is prominent near the stretching surface of the cylinder. As the distance from the surface increases, the temperature starts to drop. Moreover, thickness of thermal boundary layer is more prominent in case of stretching flat surface ($\gamma = 0.0$).

However, the temperature effects last for larger distance in case of stretching cylinder. Fig. 5 indicates that the an increase in the value of temperature exponent n results in decrease in fluid temperature $\theta(\eta)$.

The effects of the Schmidt number on the concentration are displayed in Fig.6. The Schmidt number Sc is the ratio of a viscous diffusion rate to a molecular diffusion rate. From this figure it is observed that the enhancement of Schmidt number leads to decrease the concentration of reactants.

To validate the results obtained by the applied numerical scheme, the computed numerical values for the skin friction coefficient and heat transfer coefficient are compared with some previous studies and the outcome is shown in the tables 1, 2 and 3. From table 1, it is observed that the value of $-f''(0)$ compared with that of Vajrevelu et al.[10] by taking $\gamma = 0.0$, $M = 0.0$, $K = 0.0$. An excellent agreement between the values is detected.

Table 2 is constructed to make a comparison of the values of $\theta''(0)$ with those of Grubka and Bobba [28] and Ishak and Nazar [3] for flat surface case ($\gamma = 0.0$) for several values of temperature exponent n and Prandtl number Pr when $M = 0.0$, $K = 0.0$, $Ec = 0.0$. It is observed that there is an excellent matching between the values. Finally, the values of $-f''(0)$ are compared with those of Shateyi and Marewo [12] for variation in curvature parameter by keeping $M = 0.0$, $K = 0.0$ and are presented in Table 3. The values of both the studies excellently match up to 4 decimal places. This validates the accuracy and authenticity of our numerical scheme.

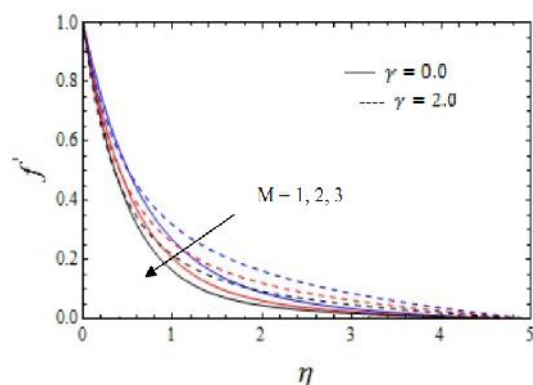


Fig.2 Effects of magnetic field parameter M on velocity profile when $K=0.5$.

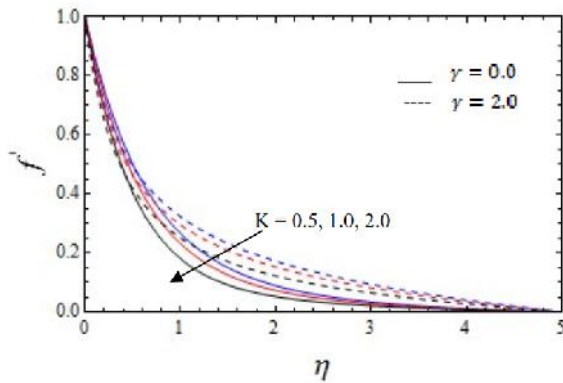


Fig.3 Effects of permeability parameter K on velocity profile when $M=1.0$.

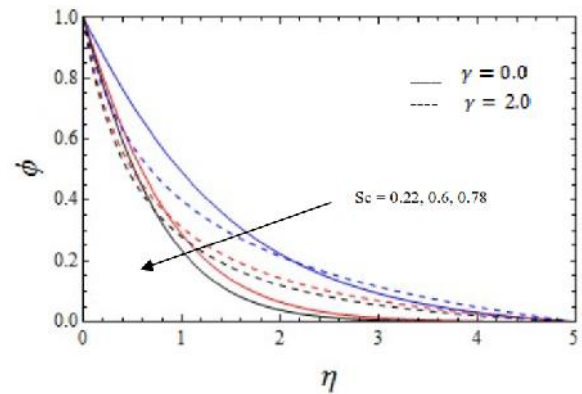


Fig.6. Effects of Schmidt number on concentration profile

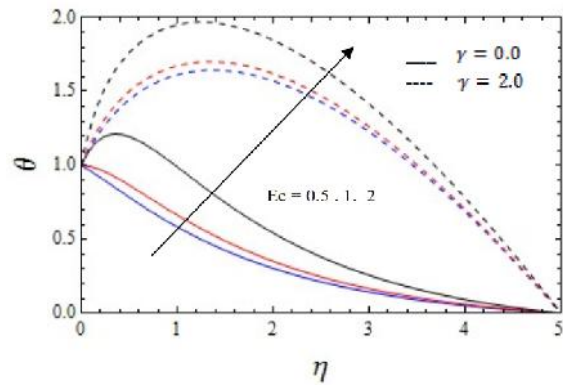


Fig.4. Effects of Eckert number Ec on temperature profile when $Pr=0.71$, $M=1$, $K=0.5$, $n=1.0$

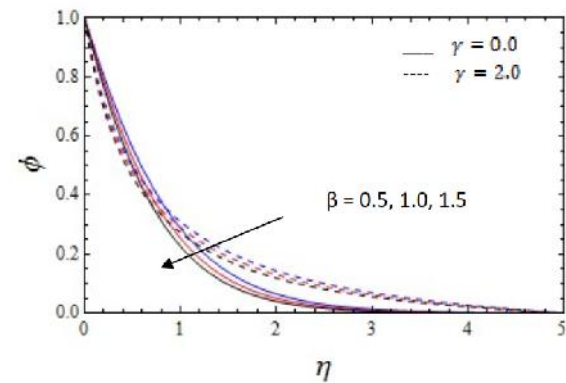


Fig.7. Effects of β on concentration profile

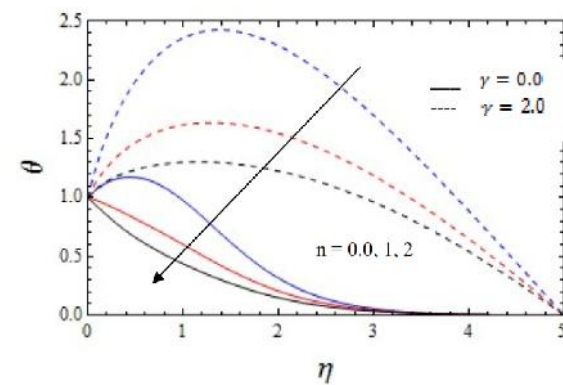


Fig.5. Effects of temperature exponent n on temperature profile when $M=1$, $K=0.5$, $Pr=0.71$, $Ec=0.5$

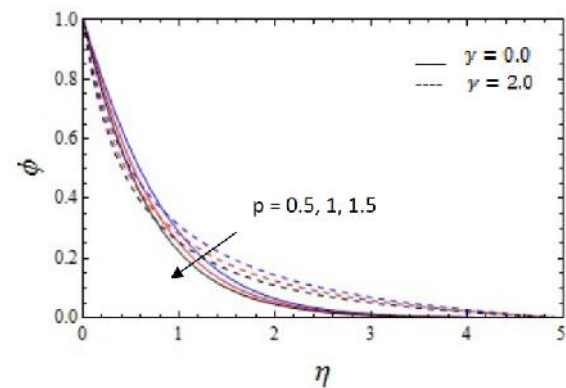


Fig.8. Effects of p on concentration profile

Table 1: Comparison of numerical value of $-f''(0)$ with that of Vajravelu [10] when $\gamma = 0.0$, $M = 0.0$, $K = 0.0$.

$-f''(0)$		$-f''(0)$
Vajravelu et al. [10]		Present
Analytical	Numerical	
1.000180	1.00000	1.005794

Table 2: Comparison of numerical values of $\theta''(0)$ for variation in Prandtl number Pr and temperature exponent n with those of Ishak and Nazar [3] and Grubka and Bobba[30] for flat surface (i.e. $\gamma = 0.0$) when $M = 0.0$, $K = 0.0$, $Ec = 0.0$.

Pr	n	$\theta''(0)$	$\theta''(0)$	$\theta''(0)$
		Ishak and Nazar[3]	Grubka and Bobba [30]	Present
1.0	-1.0	0.000	0.0000	0.0000
	0.0	0.5820	0.5820	0.5112
	1.0	1.0000	1.0000	1.0181
10.0	-1.0	0.0000	0.0000	0.0000
	0.0	2.3080	2.3080	2.3398
	1.0	3.7207	3.7207	3.7562

Table 3: Comparison of numerical values of $-f''(0)$ for variation in curvature parameter γ with those of Shateyi and Marewo [12] when $M = 0.0$, $K = 1.0$.

γ	$-f''(0)$	$-f''(0)$
	Shateyi and Marewo [12]	Present
0.0	1.4142	1.4198
0.25	1.5231	1.5154
0.5	1.6264	1.6141

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